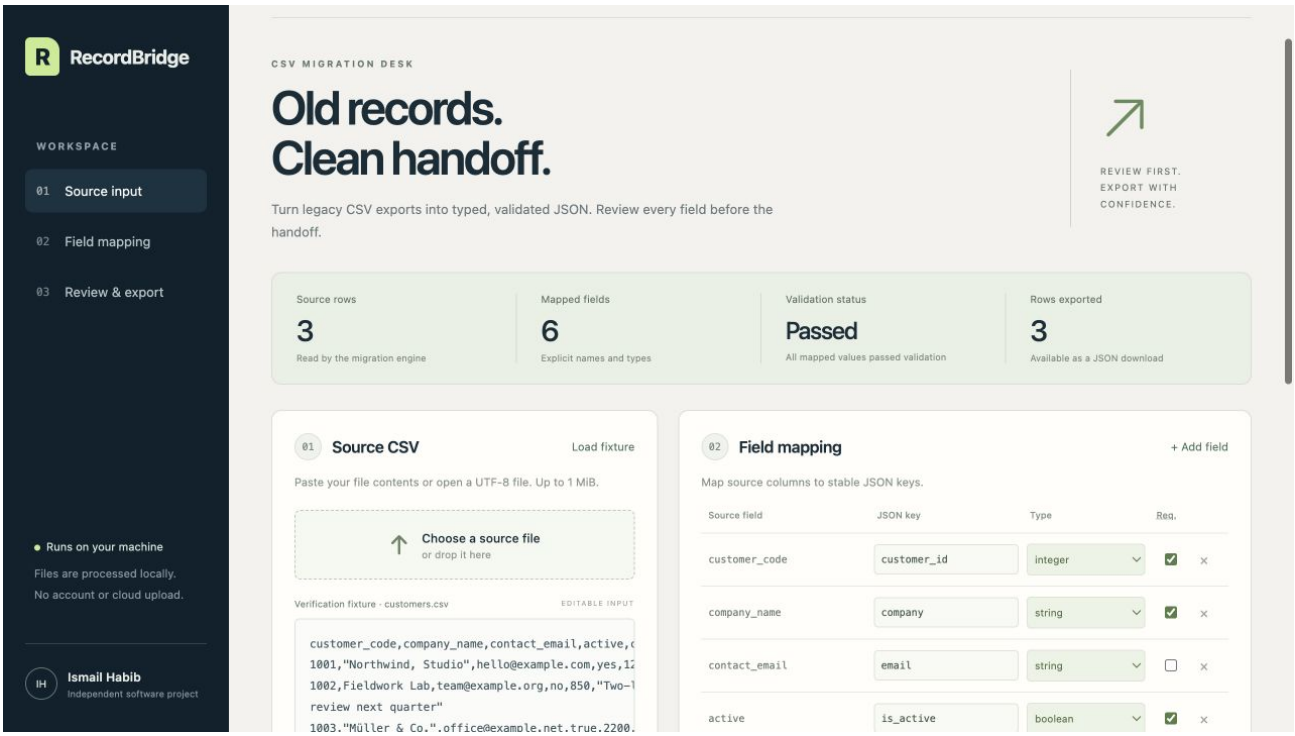
PROJECT 05 / LEGACY SOFTWARE MODERNIZATION

RecordBridge

Validated CSV-to-JSON migration with explicit field mapping

Independent software project | Owner: Ismail Habib | Created September 2026



Actual application capture. Any example records are labeled verification inputs.

The problem

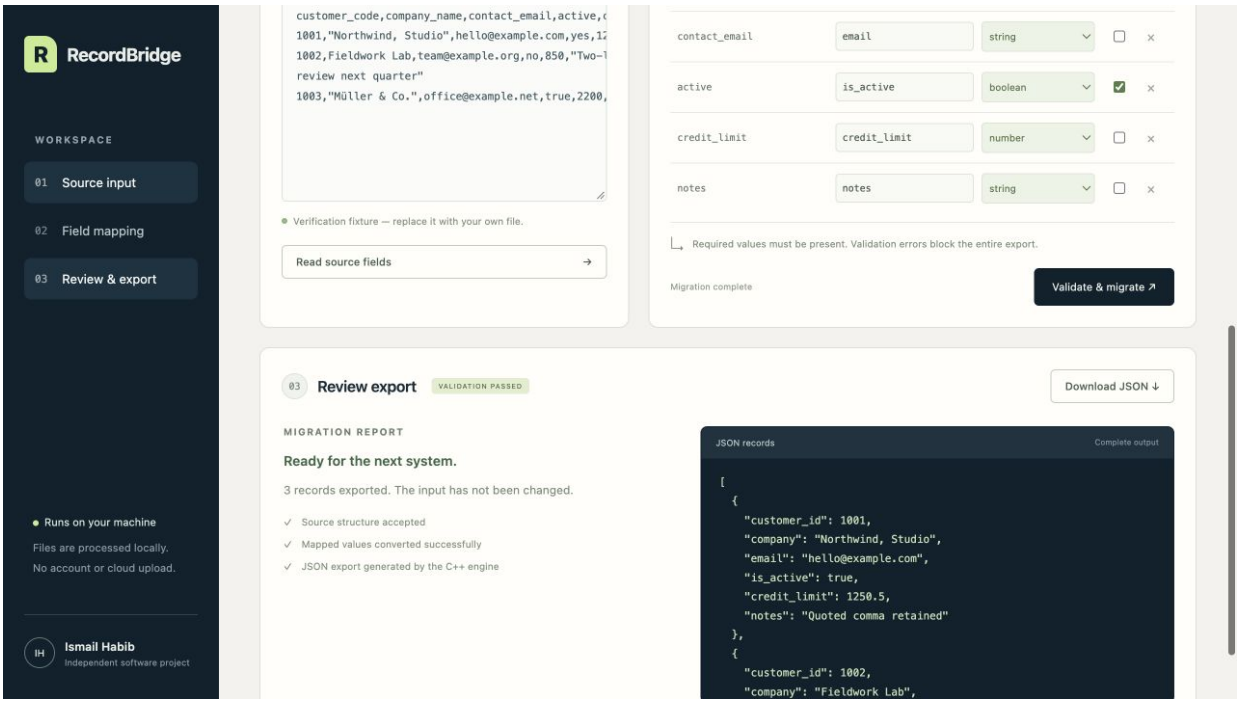
CSV exports often form the handoff between an existing application and a newer system. Quoted commas, multiline notes, duplicate headers and inconsistent row widths make naive conversion unreliable. A migration workflow must preserve source values while making field names, required values, data types and invalid records explicit before producing output.

What the project delivers

- C++17 CSV parser with quoted, multiline and Unicode field support
- Explicit field mapping with required values and declared JSON types
- Whole-export validation with source-field and line diagnostics
- Browser workflow and command-line engine that preserve originals

ENGINEERING & EVIDENCE

Built to be inspected.



The engine-generated validation report and JSON preview show the typed output produced from the actual verification input.

What was verified

Sixteen tests passed against the real compiled engine and Python adapter on 28 September 2026. The labeled fixture generated three typed JSON records while preserving quoted commas, multiline notes, Unicode and escaped quotes. Tests covered invalid types, malformed CSV, duplicate names, missing fields, invalid UTF-8 and prevention of input/output overwrite. Compilation completed without warnings using Apple Clang 17 and C++17 on macOS 15.7.8. Python and JavaScript syntax checks passed. These are local software-verification results, not measured business or customer outcomes.

How it works

Read source CSV > Map fields and types > Validate every record > Export typed JSON

Current scope

RecordBridge is a newly built migration utility supporting modernization work; it is not a migration of an existing customer's C++ Windows application. Compilation and execution were verified on macOS, with Windows behavior still unverified. Input-size, row-count and dashboard-execution limits apply. Validation enforces declared structure and types, but does not infer business rules, validate email addresses, establish identifier uniqueness or guarantee compatibility with a destination system. Real migrations need an agreed target schema and representative source-data checks before adopting the export.

Included: source code, setup guide, tests, screenshots, editable case study and Malt copy.
MIT License | Copyright 2026 Ismail Habib.